BullSO 2026 Invitational

Circuit Lab B Answer Key

1. B
2. C
3. C
4. D
5. B
6. A
7. A, D
8. C
9. D
10. B
11. A
12. B
13. D
14. C
15. A
16. B
17. B, D
18. B
19. D
20. D
21. B
22. A
23. D
24. C
25. C
26. D
27. D
28. B
29. B, C
30. A
31. B
32. C
33. C
34. B

PART 2: Short Answer

1.
   a. Resistor
   b. Capacitor
   c. Battery
   d. Diode
   e. Switch or open switch

2.
   a. 0.0018 Amps (A) or 1.8 milliamps (mA)
   b. 0.0162 Watts (W) or 16.2 milliwatts (mW)
   c. 3000 Ohms or 3k Ohms

3. 5.5 Ohms
4. 27 Ohms
5.
   a. Kirchoff's Current Law (KCL)
   b. Ohm's Law
   c. I_1 = 2.625 Amps (A), I_2 = 0.625 Amps (A)
      - -1 pt if units are missing/incorrect

6. V_1 = 14.4 Volts (V), V_2 = 17.6 Volts (V)
   - +1 pt for correct KVL equation (may be written with Ohm's Law substituted in)
   - +1 pt for correct current value
   - +1 pt for **each** correct final voltage value
   - Max final score of 3 pts if one/both final voltage values have missing/incorrect units
   - Example Solution:
     KVL: $-20 + V_1 - 12 + V_2 = 0 \rightarrow V_1 + V_2 = 32$
     Ohm's Law: $I(22) + I(18) = 32 \rightarrow 40I = 32 \rightarrow I = 0.8$ Amps (A)

V_1 = 0.8(18) = 14.4 Volts (V), V_2 = 0.8(22) = 17.6 Volts (V)

7. V_1 = 28 Volts (V), V_2 = 30 Volts (V)
   - +1 pt for **each** correct KCL loop equation
   - +1 pt for **each** correct final node voltage value
   - Max final score of 3 pts if one/both final node voltage values have missing/incorrect units
   - Example Solution:

     KCL Node 1: $\frac{V_1}{4} - \frac{V_2 - V_1}{2} = 6 \rightarrow 3V_1 - 2V_2 = 24$

     KCL Node 2: $\frac{V_2}{6} + \frac{V_2 - V_1}{2} = 6 \rightarrow -3V_1 + 4V_2 = 36$

     Solve for V_1, V_2

8. I_1 = 5 Amps (A), I_2 = 2.5 Amps (A)
   - +1 pt for **each** correct KVL loop equation
   - +1 pt for **each** correct final current value
   - Max final score of 3 pts if one/both final current values have missing/incorrect units
   - Example Solution:

     KVL Loop 1: $-25 + 4I_1 + 2(I_1 - I_2) = 0 \rightarrow 6I_1 - 2I_2 = 25$

     KVL Loop 2: $-15 + 8I_2 + 2(I_2 - I_1) = 0 \rightarrow -2I_1 + 10I_2 = 25$

     Solve for I_1, I_2

9. Any 5 of the following are acceptable

- Infinite Open-Loop Gain (G)
- Infinite Input Impedance (Rin)
- Infinite Out Voltage Range (vout max)
- Infinite Common-Mode Rejection Ratio
- Infinite Power Supply Reject Ratio
- Infinite Bandwidth
- Zero Input Offset Voltage
- Zero Output Impedance (Rout)
- Zero Noise
- Zero Input Current (vin/Rin)

Lab - answers 24 pts - 5 extra points for B given for the total and explanation

A clearly labeled sketch of your circuit. - 8 pts
Make sure they label resistor values - 3pts. Use parallel - 1.5pts. Use series - 1.5 pts.
Proper diagram - resistor images/wire connections - 1 pts. And battery/complete circuit - 1pt

Step-by-step calculations showing intermediate equivalent resistances - 8 pts
2 pts per calculation for at least 3. 2pts for final calculation

A short explanation of why your configuration produces 133.333 Ω. - 8 pts [13pts for B]
Make sure their total is 133.333 - 6 points [9 for B]. 2 points for explanation [4 for B].

- Example { series 100, 3 way parallel of 300, 50, and 150}

BullSO, University of South Florida - Repaired practice edition - Answer key

Division B | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (4 points) Authored written substitute for unavailable source material: Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Different cases can expose omitted inversions or incorrect branch logic. (4 points).